Higher-Dimensional Local Euler Characteristic Computation via Dual Cell Valuations on Regular Tessellations and Lattices

Formal Notes on Digital Topology, Cell Groups, and Valuation Theory

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Abstract

In his 1971 paper *Local Properties of Binary Images in Two Dimensions*, Stephen B. Gray established that the topological Euler characteristic of digitized binary patterns is strictly *locally countable* using neighborhood configurations termed *cell groups*. In this paper, we develop a comprehensive mathematical formalization of Gray's cell-group approach and generalize it to arbitrary regular planar tessellations (p, q), uniform cellular complexes, and N-dimensional hypercubic lattices $\mathbb{Z}^N$. We formalize: (i) continuous binary topology, the additive valuation property of the Euler characteristic, and sweep-line Morse–Bott critical point index theorems; (ii) the geometry of dual cell groups on the three regular planar lattices $(6, 3)$, $(4, 4)$, and $(3, 6)$, providing a rigorous proof of linear separability and the origin of diagonal connectivity ambiguities; (iii) exact fractional cell allocations for bit quads and symmetric triangle configurations; (iv) local Euler differentials (E^*, W^*, Z^*, H^*), simple point characterizations, and subfield decomposition theorems for topology-preserving parallel skeletonization; (v) the N-dimensional dual Voronoi valuation theorem and its equivalence to Forman's discrete Morse spherical link formula; (vi) 3-dimensional volume processing on $\mathbb{Z}^3$, including the complete classification of all 256 bit octocubes under octahedral symmetry O_h into 22 equivalence classes; and (vii) the computational complexity of local Euler counters characterized by the tuple (D, G, R, M), establishing universal diameter-2 sufficiency.

Contents

1 Introduction and Historical Context **3**

2 Continuous Binary Images, Valuations, and Morse Sweeps **3**
2.1 Continuous Definitions and Jordan Parity 3
2.2 Triangulation and the Euler–Poincaré Formula 4
2.3 The Additive Valuation Property 4
2.4 Morse–Bott Critical Point Sweeps 5

3 Planar Regular Tessellations and the Geometry of Cell Groups **5**
3.1 Classification of Regular Planar Honeycombs (p, q) 5
3.2 Formal Definition of Dual Cell Groups 5
3.3 Linear Separability of Cell Groups 6

4 Euler Characteristic Computation on 2D Regular Lattices via Cell Groups **6**
4.1 The Hexagonal Lattice $(6, 3)$ 6
4.1.1 Multiplicity Parameters and Fractional Cell Allocations 7
4.2 The Square Lattice $(4, 4)$ 8
4.2.1 Multiplicity Parameters for $\mathbb{Z}^2$ 8
4.3 The Triangular Lattice $(3, 6)$ 9

5 Euler Differentials, Simple Points, and Parallel Thinning **9**
5.1 Definition and Locality of the Euler Differential 9
5.2 Hexagonal Euler Differential Formula 9
5.3 Square Lattice Euler Differentials 10
5.4 Subfield Decomposition and Parallel Skeletonization 10

6 N-Dimensional Dual Voronoi Valuation Framework **10**
6.1 Geometric and Topological Setup 10
6.2 The Dual Voronoi Partition 11
6.3 The Universal N-Dimensional Local Valuation Theorem 11

7 Specialization to 3-Dimensional Volume Processing on $\mathbb{Z}^3$ **12**
7.1 Combinatorial Parameters and Denominator Clearing 12
7.2 Complete Classification of the 22 Octocube Equivalence Classes 12

8 Theory of Local Computability for the Euler Characteristic **13**
8.1 The Local Property Counter (LPC) Formalism 13
8.2 Fundamental Complexity Theorems for Local Euler Counting 13
8.3 Dimensional Scaling and Master Comparison Table 14

9 Conclusion **14**

1 Introduction and Historical Context

The computation of topological invariants from local data is one of the most intriguing problems in computational topology, discrete geometry, and pattern recognition. In their foundational 1969 monograph *Perceptrons*, Minsky and Papert [2] proved that *connectedness* is a global topological property that cannot be computed by diameter-limited or order-limited linear threshold devices. That is, determining whether all active bits in a binary image belong to a single connected component requires access to the entire spatial domain simultaneously.

In 1971, Stephen B. Gray published a landmark paper, *Local Properties of Binary Images in Two Dimensions* [1], demonstrating that while connectedness is non-local, the *Euler characteristic* (or Euler number):

$$\chi = B - L \tag{1}$$

—defined in two dimensions as the number of connected components (objects B) minus the number of enclosed cavities (holes L)—is strictly *locally countable*. Specifically, χ can be evaluated by sweeping a small, translation-invariant window over the digital lattice, evaluating a local function of the binary pattern, and accumulating the results.

Gray based his discrete derivations on geometric configurations he termed **cell groups**—polygons formed by connecting the centers of adjacent bit cells meeting at a grid vertex. By analyzing the fractional contributions of vertices, edges, and faces within each cell group, Gray derived explicit formulas for the hexagonal $(6, 3)$ and square $(4, 4)$ lattices, addressed the diagonal connectivity ambiguities inherent in the square grid, introduced the concept of the **Euler differential** $E^*(U) = \Delta E/\Delta U$ for parallel skeletonization, and characterized local measurements through the **Local Property Counter** (LPC) framework.

The primary objective of this monograph is to provide a rigorous, self-contained mathematical formalization of Gray's cell-group methodology and to establish its generalization to arbitrary regular tessellations (p, q), higher-dimensional cell complexes, and N-dimensional lattices $\mathbb{Z}^N$. We connect Gray's combinatorial results with modern valuation theory [3], differential Morse theory [4, 5], digital topology [7, 8], and Forman's discrete Morse theory on cellular complexes [6].

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2 Continuous Binary Images, Valuations, and Morse Sweeps

We begin by establishing the topological foundations for continuous binary patterns in $\mathbb{R}^2$ before discretizing onto cellular lattices.

2.1 Continuous Definitions and Jordan Parity

Definition 2.1 (Border Curves and Border Sets)**.** A *border curve* $\gamma \subset \mathbb{R}^2$ is the image of a simple closed Jordan curve, i.e., an injective continuous map $f : S^1 \to \mathbb{R}^2$. A *border set* $\mathcal{B} = \{\gamma_1, \ldots, \gamma_m\}$ is a finite collection of pairwise disjoint border curves:

$$\gamma_i \cap \gamma_j = \emptyset \quad \forall i \neq j. \tag{2}$$

By the Jordan Curve Theorem, each border curve γ_i partitions the plane $\mathbb{R}^2 \setminus \gamma_i$ into two connected components: a bounded interior $\mathrm{Int}(\gamma_i)$ and an unbounded exterior $\mathrm{Ext}(\gamma_i)$.

Definition 2.2 (Parity Function and Continuous Binary Image)**.** Let $\mathcal{B} = \{\gamma_1, \ldots, \gamma_m\}$ be a border set. For any point $x \in \mathbb{R}^2 \setminus \bigcup_{i=1}^m \gamma_i$, let $\rho_x : [0, \infty) \to \mathbb{R}^2$ be a smooth ray from $\rho_x(0) = x$ to infinity intersecting $\bigcup \gamma_i$ transversally. The *parity* $P(x) \in \{0, 1\}$ is:

$$P(x) \coloneqq \left| \rho_x([0, \infty)) \cap \bigcup_{i=1}^{m} \gamma_i \right| \pmod 2. \tag{3}$$

The induced *binary image* $I : \mathbb{R}^2 \to \{0, 1\}$ sets $I(x) = P(x)$ on regular points and 0 on curves. The *foreground* is $X \coloneqq \{x \in \mathbb{R}^2 \mid I(x) = 1\}$, and the *background* is $X^c \coloneqq \mathbb{R}^2 \setminus X$.

- An *object* is a connected component of X. The number of objects is $B = b_0(X) = \dim H_0(X; \mathbb{R})$.
- A *hole* is a bounded connected component of X^c. The number of holes is $L = b_1(X) = \dim H_1(X; \mathbb{R})$.

Proposition 2.3 (Parity Well-Definedness)**.** *$P(x)$ is independent of the choice of transverse ray ρ_x, and is constant on each connected component of $\mathbb{R}^2 \setminus \bigcup \gamma_i$.*

2.2 Triangulation and the Euler–Poincaré Formula

Definition 2.4 (Continuous Image Triangulation)**.** Let $X \subset \mathbb{R}^2$ be a compact binary foreground. A *triangulation* $\mathcal{K}$ of X is a finite simplicial 2-complex homeomorphic to $\bar{X}$ such that every 2-simplex $\sigma_2 \in \mathcal{C}_2(\mathcal{K})$ has $I(x) = 1$ on its interior, and the boundary curves ∂X form a subcomplex.

Theorem 2.5 (Euler–Poincaré Theorem for Planar Binary Images)**.** *Let n_0, n_1, n_2 denote the total number of 0-simplices (vertices), 1-simplices (edges), and 2-simplices (faces) in any valid triangulation $\mathcal{K}$ of X. Then:*

$$\chi(X) = n_0 - n_1 + n_2 = B - L. \tag{4}$$

The quantity $\chi(X)$ is an intrinsic topological invariant, independent of the triangulation chosen.

Proof. By the Euler–Poincaré theorem for finite cell complexes:

$$n_0 - n_1 + n_2 = \sum_{k=0}^{2} (-1)^k \dim H_k(X; \mathbb{R}).$$

For a compact 2D region in the plane, $H_0(X; \mathbb{R}) \cong \mathbb{R}^B$, $H_1(X; \mathbb{R}) \cong \mathbb{R}^L$, and $H_2(X; \mathbb{R}) \cong 0$. Thus $\sum_{k=0}^{2} (-1)^k \dim H_k(X; \mathbb{R}) = B - L$. □

2.3 The Additive Valuation Property

Theorem 2.6 (Valuation Property of Euler Characteristic)**.** *Let $I_1, I_2 \subset \mathbb{R}^2$ be two compact, finitely triangulable binary foregrounds. The Euler characteristic satisfies the inclusion-exclusion valuation identity:*

$$\chi(I_1 \cup I_2) = \chi(I_1) + \chi(I_2) - \chi(I_1 \cap I_2). \tag{5}$$

Proof. Let $\mathcal{K}$ be a common triangulation for $I_1 \cup I_2$, I_1, I_2, and $I_1 \cap I_2$. For each dimension $k \in \{0, 1, 2\}$, the cell counts satisfy $n_k(I_1 \cup I_2) = n_k(I_1) + n_k(I_2) - n_k(I_1 \cap I_2)$. Taking the alternating sum $\sum_{k=0}^{2} (-1)^k n_k$ yields the result. □

Corollary 2.7 (Inclusion-Exclusion for N Overlapping Regions)**.** *For any finite collection of binary images $\{I_1, I_2, \ldots, I_N\}$:*

$$\chi\left(\bigcup_{i=1}^{N} I_i\right) = \sum_{k=1}^{N} (-1)^{k-1} \sum_{1 \le i_1 < \cdots < i_k \le N} \chi\left(I_{i_1} \cap \cdots \cap I_{i_k}\right). \tag{6}$$

Lemma 2.8 (Linearity Condition for Adding Disjoint Regions)**.** *Let I be a base image, and let A, B be two modifications added to I. Let $\Delta\chi_A \coloneqq \chi(I \cup A) - \chi(I)$ and $\Delta\chi_B \coloneqq \chi(I \cup B) - \chi(I)$. Then:*

$$\Delta\chi_{A \cup B} - \Delta\chi_A - \Delta\chi_B = \chi(I \cap A \cap B) - \chi(A \cap B). \tag{7}$$

In particular, if $A \cap B = \emptyset$, then the additions are strictly linear:

$$\Delta\chi_{A \cup B} = \Delta\chi_A + \Delta\chi_B. \tag{8}$$

Proof. Applying Theorem 2.6 to $(I \cup A) \cup (I \cup B) = I \cup A \cup B$:

$$\begin{aligned}\chi(I \cup A \cup B) &= \chi(I \cup A) + \chi(I \cup B) - \chi((I \cup A) \cap (I \cup B)) \\ &= \chi(I \cup A) + \chi(I \cup B) - \chi(I \cup (A \cap B)).\end{aligned}$$

Subtracting $\chi(I)$ and expanding $\chi(I \cup (A \cap B)) = \chi(I) + \chi(A \cap B) - \chi(I \cap A \cap B)$ yields the result. When $A \cap B = \emptyset$, both terms on the right are zero. □

2.4 Morse–Bott Critical Point Sweeps

Gray's critical point method can be formalized through continuous Morse theory [4, 5].

Definition 2.9 (Sweep Line and Height Function)**.** Let $\theta \in S^1$ be a unit direction vector in $\mathbb{R}^2$. The linear height function is $h_\theta(x) \coloneqq \langle x, \theta \rangle$. For $t \in \mathbb{R}$, let $X_t \coloneqq X \cap \{x \in \mathbb{R}^2 \mid h_\theta(x) \leq t\}$ be the sub-level set.

Theorem 2.10 (Morse Index Theorem for Binary Images)**.** *Assume ∂X is smooth with non-degenerate critical points with respect to $h_\theta|_{\partial X}$. The Euler characteristic changes only when t passes through a critical point:*

1. ***Upstream Convexity** (X_θ): Index 0 critical point (local minimum of $h_\theta|_{\partial X}$ with outward normal opposing θ). A new component is created: $\Delta\chi = +1$.*
2. ***Upstream Concavity** (V_θ): Index 1 critical point (saddle/local minimum of the complementary domain). Two components merge or a hole is formed: $\Delta\chi = -1$.*

Consequently, summing all critical points along direction θ yields:

$$\chi(X) = B - L = n(X_\theta) - n(V_\theta). \tag{9}$$

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3 Planar Regular Tessellations and the Geometry of Cell Groups

3.1 Classification of Regular Planar Honeycombs (p, q)

Definition 3.1 (Regular Planar Tessellation (p, q))**.** A *regular tessellation* of the Euclidean plane $\mathbb{R}^2$, denoted by the Schläfli symbol (p, q), is a partition of $\mathbb{R}^2$ into regular p-sided convex polygons (termed **bit cells**) such that exactly q polygons meet at each vertex.

Proposition 3.2 (The Three Regular Lattices)**.** *The only regular tessellations of the Euclidean plane are:*

$$(6, 3) \text{ (Hexagonal)}, \quad (4, 4) \text{ (Square)}, \quad (3, 6) \text{ (Triangular)}. \tag{10}$$

Proof. In a regular Euclidean p-gon, the interior vertex angle is $\alpha_p = \frac{(p-2)\pi}{p}$. For q such polygons to surround a vertex without gaps, $q \cdot \alpha_p = 2\pi \iff (p-2)(q-2) = 4$. For integers $p, q \geq 3$, the only factorizations of 4 are $(4, 1), (2, 2), (1, 4)$, corresponding to $(p, q) \in \{(6, 3), (4, 4), (3, 6)\}$. □

3.2 Formal Definition of Dual Cell Groups

Definition 3.3 (Primal Vertices and Dual Cell Groups)**.** Let $\mathcal{T} = (p, q)$ be a regular planar tessellation.

- Let $\mathcal{C}_2(\mathcal{T})$ denote the set of open 2-cells (primal bit cells).
- Let $\mathcal{V}(\mathcal{T}) = \mathcal{C}_0(\mathcal{T})$ denote the set of primal 0-cells (grid vertices).

- For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, exactly q bit cells $\{\sigma_2^{(1)}, \ldots, \sigma_2^{(q)}\}$ meet at v. The **cell group** $G(v)$ is the convex q-gon formed by connecting the geometric centers of these q bit cells in cyclic order:

$$G(v) \coloneqq \operatorname{conv}\left(\operatorname{center}(\sigma_2^{(1)}), \operatorname{center}(\sigma_2^{(2)}), \ldots, \operatorname{center}(\sigma_2^{(q)})\right). \tag{11}$$

Proposition 3.4 (Dual Voronoi Property of Cell Groups)**.** *The collection of cell groups $\{G(v)\}_{v\in\mathcal{V}(\mathcal{T})}$ forms the dual tessellation (q, p) of the plane. Each cell group $G(v)$ is geometrically centered at the primal vertex v.*

3.3 Linear Separability of Cell Groups

When a continuous binary image with smooth border curves is digitized onto (p, q), each bit cell takes state 1 or 0 depending on whether its center lies in the foreground.

Definition 3.5 (Linear Separability)**.** A cell group configuration $(b_1, \ldots, b_q) \in \{0, 1\}^q$ is *linearly separable* if there exists a straight line in the plane separating the vertices with $b_i = 1$ from those with $b_i = 0$.

Theorem 3.6 (Separability Comparison Across Planar Lattices)**.**

1. ***Hexagonal Lattice*** $(6, 3)$*: Cell groups are triangles ($q = 3$). In $\mathbb{R}^2$,* every *non-trivial partition of the 3 vertices of a triangle into $\{0\}$ and $\{1\}$ is linearly separable. Thus, the inseparability count is* ***0 out of 8*** *(0%). Diagonal connectivity ambiguities are geometrically impossible.*

2. ***Square Lattice*** $(4, 4)$*: Cell groups are squares ($q = 4$). Exactly 2 of the 16 binary patterns (12.5%)—namely the diagonal configurations* $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ *and* $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ *(denoted Q_D)—are non-linearly separable (the XOR pattern).*

3. ***Triangular Lattice*** $(3, 6)$*: Cell groups are hexagons ($q = 6$). Exactly 32 of the 64 binary patterns (50%) are non-linearly separable.*

Proof. For $q = 3$, let $S \subset \{v_1, v_2, v_3\}$ be the set of 1-vertices. If $|S| \in \{1, 2\}$, $\operatorname{conv}(S)$ is either a point or a line segment, and $\operatorname{conv}(\{v_1, v_2, v_3\} \setminus S)$ is a segment or a point disjoint from $\operatorname{conv}(S)$. By the Hyperplane Separation Theorem, they are strictly linearly separable.

For $q = 4$, the diagonals of a convex quad intersect at its interior center point $(1/2, 1/2)$. For the diagonal patterns Q_D, $\operatorname{conv}(v_1, v_3) \cap \operatorname{conv}(v_2, v_4) = \{(1/2, 1/2)\} \neq \emptyset$, precluding linear separation.

For $q = 6$, any alternating binary pattern $(1, 0, 1, 0, 1, 0)$ or pattern with 3 disjoint 1-vertices has intersecting convex hulls, resulting in 32 inseparable configurations. □

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4 Euler Characteristic Computation on 2D Regular Lattices via Cell Groups

We now derive the explicit combinatorial Euler characteristic formulas for the three regular planar lattices by decomposing the cellular complex across dual cell groups.

4.1 The Hexagonal Lattice $(6, 3)$

On the hexagonal lattice Λ_{hex}, bit cells are regular hexagons ($p = 6$) and cell groups are equilateral triangles ($q = 3$).

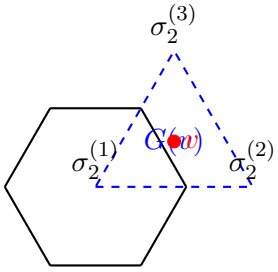


Figure 1: The hexagonal cell group $G(v)$ is an equilateral triangle ($q = 3$) centered at primal vertex v, connecting the centers of the three incident hexagons.

4.1.1 Multiplicity Parameters and Fractional Cell Allocations

- $V_0 = 1$: Each 0-cell (vertex) is shared by 1 dual cell group.
- $V_1 = 2$: Each 1-cell (hexagon edge) has 2 vertices, hence is shared by $V_1 = 2$ dual triangles.
- $V_2 = 6$: Each 2-cell (hexagon) has 6 vertices, hence is shared by $V_2 = 6$ dual triangles.

Theorem 4.1 (Directional Critical Point Formula for Hexagons)**.** *Choosing a sweep direction θ parallel to a hexagon edge, the critical point counts evaluate to:*

$$n(X_\theta) = n \begin{bmatrix} & 1 & \\ 0 & & 0 \end{bmatrix}, \qquad n(V_\theta) = n \begin{bmatrix} & 0 & \\ 1 & & 1 \end{bmatrix}. \tag{12}$$

By Theorem 2.10, the discrete Euler characteristic is:

$$E = n \begin{bmatrix} & 1 & \\ 0 & & 0 \end{bmatrix} - n \begin{bmatrix} & 0 & \\ 1 & & 1 \end{bmatrix}. \tag{13}$$

Theorem 4.2 (Isotropic Symmetric Hexagonal Triangle Formula)**.** *Let T_1 denote the total number of cell group triangles with exactly one* 1 *and two* 0*s, and let T_2 denote the total number of cell group triangles with exactly two* 1*s and one* 0*. Then:*

$$E = \frac{T_1 - T_2}{6}. \tag{14}$$

Proof. Let $Q(v) = (b_1, b_2, b_3) \in \{0,1\}^3$ denote the binary state of the cell group triangle $G(v)$. The fractional cell contributions $\Delta n_k(Q(v))$ are:

- If $Q(v) = (0,0,0)$: $\Delta n_0 = 0, \Delta n_1 = 0, \Delta n_2 = 0 \implies \Delta\chi = 0$.
- If $Q(v)$ has one 1 (T_1 type): $\Delta n_0 = 1$, $\Delta n_1 = 2 \times \frac{1}{2} = 1$, $\Delta n_2 = 1 \times \frac{1}{6} = \frac{1}{6}$. Thus:
$$\Delta\chi(T_1) = 1 - 1 + \frac{1}{6} = +\frac{1}{6}.$$
- If $Q(v)$ has two 1s (T_2 type): $\Delta n_0 = 1$, $\Delta n_1 = 3 \times \frac{1}{2} = \frac{3}{2}$, $\Delta n_2 = 2 \times \frac{1}{6} = \frac{1}{3}$. Thus:
$$\Delta\chi(T_2) = 1 - \frac{3}{2} + \frac{1}{3} = -\frac{1}{6}.$$
- If $Q(v) = (1,1,1)$ (T_3 type): $\Delta n_0 = 1$, $\Delta n_1 = 3 \times \frac{1}{2} = \frac{3}{2}$, $\Delta n_2 = 3 \times \frac{1}{6} = \frac{1}{2}$. Thus:
$$\Delta\chi(T_3) = 1 - \frac{3}{2} + \frac{1}{2} = 0.$$

Summing over all grid vertices $v \in \mathcal{V}(\mathcal{T})$:

$$E = \sum_{v \in \mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)) = \frac{1}{6}T_1 - \frac{1}{6}T_2 + 0T_3 = \frac{T_1 - T_2}{6}. \qquad \square$$

4.2 The Square Lattice $(4, 4)$

On the square grid $\mathbb{Z}^2$, bit cells are unit squares ($p = 4$) and cell groups are squares ($q = 4$) connecting the 4 centers of bit cells meeting at grid intersection v.

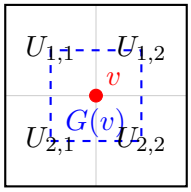


Figure 2: The square cell group $G(v)$ is a 2×2 bit quad centered at primal vertex v.

4.2.1 Multiplicity Parameters for $\mathbb{Z}^2$

- $V_0 = 1$ (1 vertex per 0-cell).
- $V_1 = 2$ (2 vertices per unit edge).
- $V_2 = 4$ (4 vertices per unit square).

A 2×2 window contains $2^4 = 16$ configurations, partitioned into 6 rotation equivalence classes:

$$Q_0\left(\begin{bmatrix}0&0\\0&0\end{bmatrix}\right),\quad Q_1\left(\begin{bmatrix}1&0\\0&0\end{bmatrix}\right),\quad Q_2\left(\begin{bmatrix}1&1\\0&0\end{bmatrix}\right),\quad Q_D\left(\begin{bmatrix}1&0\\0&1\end{bmatrix},\begin{bmatrix}0&1\\1&0\end{bmatrix}\right),\quad Q_3\left(\begin{bmatrix}1&1\\1&0\end{bmatrix}\right),\quad Q_4\left(\begin{bmatrix}1&1\\1&1\end{bmatrix}\right). \tag{15}$$

Table 1: Complete fractional cell allocations and local Euler valuations for square lattice bit quads.

Quad Class	**Multiplicity**	Δn_0	Δn_1	Δn_2	$\Delta\chi_W$ (8-conn)	$\Delta\chi_Z$ (4-conn)
Q_0	1	0	0	0	0	0
Q_1	4	1	$2 \times \frac{1}{2} = 1$	$1 \times \frac{1}{4} = \frac{1}{4}$	$1 - 1 + \frac{1}{4} = +\frac{1}{4}$	$+\frac{1}{4}$
Q_2	4	1	$3 \times \frac{1}{2} = \frac{3}{2}$	$2 \times \frac{1}{4} = \frac{1}{2}$	$1 - \frac{3}{2} + \frac{1}{2} = 0$	0
Q_D (8-conn)	2	1	$4 \times \frac{1}{2} = 2$	$2 \times \frac{1}{4} = \frac{1}{2}$	$1 - 2 + \frac{1}{2} = -\frac{1}{2}$	–
Q_D (4-conn)	2	2	$4 \times \frac{1}{2} = 2$	$2 \times \frac{1}{4} = \frac{1}{2}$	–	$2 - 2 + \frac{1}{2} = +\frac{1}{2}$
Q_3	4	1	$3 \times \frac{1}{2} = \frac{3}{2}$	$3 \times \frac{1}{4} = \frac{3}{4}$	$1 - \frac{3}{2} + \frac{3}{4} = -\frac{1}{4}$	$-\frac{1}{4}$
Q_4	1	1	$2 \times \frac{1}{2} = 1$	$4 \times \frac{1}{4} = 1$	$1 - 1 + 1 = 0$	0

Theorem 4.3 (Square Lattice Euler Characteristic Formulas)**.** *Summing the fractional valuations over all quads in the image:*

$$4W = n(Q_1) - n(Q_3) - 2n(Q_D) \quad \textit{(8-connectivity / 1s-joined)}, \tag{16}$$

$$4Z = n(Q_1) - n(Q_3) + 2n(Q_D) \quad \textit{(4-connectivity / 0s-joined)}, \tag{17}$$

$$4H = n(Q_1) - n(Q_3) \quad \textit{(Symmetric half-connected model)}. \tag{18}$$

Proof. By Table 1, the global Euler characteristic under 8-connectivity is:

$$W = \sum_{v \in \mathbb{Z}^2} \Delta\chi_W(Q(v)) = \frac{1}{4}n(Q_1) + 0n(Q_2) - \frac{1}{2}n(Q_D) - \frac{1}{4}n(Q_3) + 0n(Q_4).$$

Multiplying by 4 gives $4W = n(Q_1) - n(Q_3) - 2n(Q_D)$. Under 4-connectivity, the two diagonal 1s meet without sharing a vertex, so $\Delta n_0 = 2$, yielding $\Delta\chi_Z(Q_D) = +1/2$. Multiplying by 4 gives $4Z = n(Q_1) - n(Q_3) + 2n(Q_D)$. The average $H = \frac{W+Z}{2}$ clears the $n(Q_D)$ term, giving $4H = n(Q_1) - n(Q_3)$. □

4.3 The Triangular Lattice $(3, 6)$

On the triangular lattice $(3, 6)$, bit cells are triangles ($p = 3$) and cell groups are regular hexagons ($q = 6$).

- Multiplicity parameters: $V_0 = 1, V_1 = 2, V_2 = 3$.
- Each cell group hexagon contains v ($\Delta n_0 = 1$), $\frac{1}{2}$ of each incident edge, and $\frac{1}{3}$ of each incident triangle.
- The local contribution formula is $\Delta\chi(Q(v)) = \Delta n_0 - \frac{1}{2}n_1^{\text{loc}} + \frac{1}{3}n_2^{\text{loc}}$.

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5 Euler Differentials, Simple Points, and Parallel Thinning

5.1 Definition and Locality of the Euler Differential

Definition 5.1 (Euler Differential)**.** Let U be a bit cell in a binary image I. The *Euler differential* $E^*(U)$ is the discrete derivative:

$$E^*(U) \coloneqq \frac{\Delta E}{\Delta U} = \begin{cases} E(I|_{U=1}) - E(I|_{U=0}) & \text{if } \Delta U = +1, \\ E(I|_{U=0}) - E(I|_{U=1}) & \text{if } \Delta U = -1. \end{cases} \tag{19}$$

Theorem 5.2 (Locality of $E^*(U)$)**.** *For a regular tessellation (p, q), toggling bit cell U modifies exactly the p cell groups centered at the p boundary vertices of U. Consequently, $E^*(U)$ is strictly a function of the immediate q-neighbor ring surrounding U.*

5.2 Hexagonal Euler Differential Formula

Let $\mathcal{N}_6(U) = (U_0, U_1, \ldots, U_5)$ be the 6 cyclic neighbors of U.

Theorem 5.3 (Hexagonal Differential Theorem)**.** *Let $n(0, 1)$ be the number of $0 \to 1$ transitions in cyclic order $(U_0, \ldots, U_5, U_0)$. Then:*

$$E^*(U) = 1 - n(0, 1) = 1 - \sum_{i=0}^{5} \bar{U}_i U_{i+1} = 1 - |U'|, \tag{20}$$

where $U'_i \coloneqq \bar{U}_i \wedge U_{i+1}$ (with indices modulo 6).

Proof. When U transitions from $0 \to 1$:

- Exactly one 2-cell is added: $\Delta n_2 = +1$.
- A unit edge is added for every 0-neighbor: $\Delta n_1 = n(0)$.
- A vertex is added for every consecutive pair of zero neighbors: $\Delta n_0 = n(0, 0)$.

Thus $\Delta E = n(0, 0) - n(0) + 1$. Using the cyclic transition identity $n(0) = n(0, 0) + n(0, 1)$, we obtain $E^*(U) = 1 - n(0, 1)$. □

Proposition 5.4 (Topological Taxonomy of $E^*(U)$)**.**

- $E^*(U) = +1$*: U creates a new isolated connected component (if $|U| = 0$) or eliminates a hole (if $|U| = 6$).*
- $E^*(U) = 0$*: U is a* ***simple point****. Deleting U preserves all topological Betti numbers.*
- $E^*(U) = -1$*: U joins two disconnected components or closes a loop (creating a hole).*
- $E^*(U) \leq -2$*: U is a multi-way branch point joining ≥ 3 components.*

5.3 Square Lattice Euler Differentials

For the 8-neighborhood $\mathcal{N}_8(U) = (U_0, \ldots, U_7)$ on $\mathbb{Z}^2$ (with even indices for edge neighbors and odd indices for corner neighbors):

Theorem 5.5 (Square Lattice Differential Formulas)**.** *Let C_{246} count corner triplets matching $\{0,1,0\}, \{1,0,0\}, \{1,1,0\}$. Let C_{135} count corner triplets matching $\{0,0,1\}, \{0,1,1\}, \{1,0,1\}$. Let C_{07} count homogeneous corners ($\{0,0,0\}$ or $\{1,1,1\}$). Then:*

$$W^*(U) = 1 - C_{246} = 1 - |U'_e|, \tag{21}$$

$$Z^*(U) = 1 - C_{135} = 1 - |U'_d|, \tag{22}$$

$$H^*(U) = \frac{1}{2}C_{07} - 1 = 1 - \frac{1}{2}|U'|. \tag{23}$$

5.4 Subfield Decomposition and Parallel Skeletonization

Theorem 5.6 (Subfield Thinning Theorem)**.** *Let Λ be a regular lattice partitioned into M disjoint sublattices $\Lambda = \bigcup_{k=1}^{M} \Lambda^{(k)}$ such that no two cells in any single sublattice $\Lambda^{(k)}$ share a cell group. Then simultaneously removing all simple points ($E^*(U) = 0$) within $\Lambda^{(k)}$ satisfies:*

$$\Delta E = \sum_{U \in \Lambda^{(k)}, E^*(U)=0} E^*(U) = 0. \tag{24}$$

The minimal number of subfields required for collision-free parallel thinning is:

$$M_{hex} = 3, \quad M_{square} = 4, \quad M_{tri} = 6, \quad M_{\mathbb{Z}^N} = 2^N. \tag{25}$$

Proof. By Theorem 2.8, if all modified cells in a batch are mutually disjoint in support ($A \cap B = \emptyset$), the joint topological valuation is strictly linear. Partitioning the lattice into independent cosets ensures all simultaneous updates are disjoint. □

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6 N-Dimensional Dual Voronoi Valuation Framework

We now generalize the cell-group valuation framework to arbitrary N-dimensional uniform and regular cellular complexes.

6.1 Geometric and Topological Setup

Let $\mathbb{M}^N$ be an N-dimensional simply connected Riemannian manifold of constant curvature ($\mathbb{R}^N$, $\mathbb{S}^N$, or $\mathbb{H}^N$).

Definition 6.1 (N-Dimensional Regular/Uniform Cellular Complex)**.** Let $\mathcal{T}$ be a locally finite, regular or uniform cellular decomposition of $\mathbb{M}^N$. For each dimension $k \in \{0, 1, \ldots, N\}$, let $\mathcal{C}_k(\mathcal{T})$ denote the set of open k-cells. The top-dimensional cells $\sigma_N \in \mathcal{C}_N(\mathcal{T})$ are termed *generalized voxels*.

Definition 6.2 (N-Dimensional Binary Image and Foreground Complex)**.** A *binary image* on $\mathcal{T}$ is a compactly supported map $I : \mathcal{C}_N(\mathcal{T}) \to \{0,1\}$. The *foreground complex* $K(I) \subseteq \mathcal{T}$ is the closed subcomplex:

$$K(I) \coloneqq \bigcup_{\substack{\sigma_N \in \mathcal{C}_N(\mathcal{T}) \\ I(\sigma_N)=1}} \bar{\sigma}_N. \tag{26}$$

The set of active k-cells is $\mathcal{C}_k(K(I)) \coloneqq \{\sigma_k \in \mathcal{C}_k(\mathcal{T}) \mid \sigma_k \subseteq K(I)\}$, and $n_k(K(I)) \coloneqq |\mathcal{C}_k(K(I))|$.

Definition 6.3 (Euler–Poincaré Characteristic)**.**

$$\chi(K(I)) = \sum_{k=0}^{N} (-1)^k n_k(K(I)). \tag{27}$$

6.2 The Dual Voronoi Partition

Let $\mathcal{V}(\mathcal{T}) = \mathcal{C}_0(\mathcal{T})$ denote the primal 0-cells of $\mathcal{T}$.

Definition 6.4 (Dual Voronoi Cell Group)**.** For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, the *dual Voronoi cell* $C(v)$ is:

$$C(v) \coloneqq \left\{ x \in \mathbb{M}^N \;\middle|\; d(x,v) \leq d(x,w) \quad \forall w \in \mathcal{V}(\mathcal{T}) \right\}. \tag{28}$$

Definition 6.5 (Vertex Multiplicity of k-Cells)**.** For a regular or uniform tessellation $\mathcal{T}$, each k-cell $\sigma_k \in \mathcal{C}_k(\mathcal{T})$ is spanned by a fixed number of boundary vertices:

$$V_k \coloneqq |\mathcal{V}(\bar{\sigma}_k)| = |\mathcal{C}_0(\bar{\sigma}_k)|. \tag{29}$$

Lemma 6.6 (Uniform Multiplicity of Cell Partitioning)**.** *Let $\{C(v)\}_{v \in \mathcal{V}(\mathcal{T})}$ be the dual Voronoi partition of $\mathbb{M}^N$. For every $k \in \{0, \ldots, N\}$ and every k-cell $\sigma_k \in \mathcal{C}_k(\mathcal{T})$:*

$$\sum_{v \in \mathcal{V}(\mathcal{T})} \mathbf{1}_{\{\sigma_k \cap \mathrm{int}(C(v)) \neq \emptyset\}} = V_k. \tag{30}$$

Proof. By vertex-transitivity and symmetry of $\mathcal{T}$:

1. For $k = 0$, v is the center of $C(v)$, so $V_0 = 1$.
2. For $k = 1$, any edge $e = (v_1, v_2)$ is bisected by $\partial C(v_1) \cap \partial C(v_2)$, intersecting exactly $V_1 = 2$ dual cells.
3. For $k \geq 2$, the barycentric subdivision of σ_k divides the interior into V_k congruent sectors, each contained in the Voronoi cell $C(v_i)$ of one of its V_k spanning vertices $v_i \in \mathcal{V}(\bar{\sigma}_k)$. □

6.3 The Universal N-Dimensional Local Valuation Theorem

Definition 6.7 (Local Neighborhood Configuration $Q(v)$)**.** For each primal vertex $v \in \mathcal{V}(\mathcal{T})$, let $\mathrm{Star}_N(v) = \{\sigma_N^{(1)}, \ldots, \sigma_N^{(D_v)}\}$ be the set of D_v distinct N-cells meeting at v. The local configuration vector is:

$$Q(v) \coloneqq \left(I(\sigma_N^{(1)}), I(\sigma_N^{(2)}), \ldots, I(\sigma_N^{(D_v)}) \right) \in \{0,1\}^{D_v}. \tag{31}$$

Theorem 6.8 (Universal N-Dimensional Local Valuation Theorem)**.** *The global Euler characteristic $\chi(K(I))$ is identically equal to the sum of local Euler valuations evaluated over all primal vertices:*

$$\chi(K(I)) = \sum_{v \in \mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)), \tag{32}$$

where:

$$\Delta\chi(Q(v)) = \sum_{k=0}^{N} (-1)^k \Delta n_k(Q(v)), \quad \textit{with} \quad \Delta n_k(Q(v)) \coloneqq \frac{1}{V_k} \cdot \left| \{\sigma_k \in \mathcal{C}_k(K(I)) \mid v \in \bar{\sigma}_k\} \right|. \tag{33}$$

Proof. Summing over all vertices $v \in \mathcal{V}(\mathcal{T})$ and swapping the summation order:

$$\begin{aligned}\sum_{v\in\mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)) &= \sum_{v\in\mathcal{V}(\mathcal{T})} \sum_{k=0}^{N} (-1)^k \Delta n_k(Q(v)) \\ &= \sum_{k=0}^{N} (-1)^k \sum_{v\in\mathcal{V}(\mathcal{T})} \frac{1}{V_k} \sum_{\substack{\sigma_k\in\mathcal{C}_k(K(I)) \\ v\in\bar{\sigma}_k}} 1 \\ &= \sum_{k=0}^{N} (-1)^k \frac{1}{V_k} \sum_{\sigma_k\in\mathcal{C}_k(K(I))} \sum_{v\in\mathcal{V}(\bar{\sigma}_k)} 1.\end{aligned}$$

By definition of V_k, the innermost sum is $\sum_{v\in\mathcal{V}(\bar{\sigma}_k)} 1 = V_k$. Therefore:

$$\sum_{v\in\mathcal{V}(\mathcal{T})} \Delta\chi(Q(v)) = \sum_{k=0}^{N} (-1)^k \frac{1}{V_k} \sum_{\sigma_k\in\mathcal{C}_k(K(I))} V_k = \sum_{k=0}^{N} (-1)^k n_k(K(I)) = \chi(K(I)). \qquad \square$$

Theorem 6.9 (Forman's Discrete Morse Spherical Link Formula)**.** *Let $S_\varepsilon^{N-1}(v)$ be a geodesic sphere of radius $\varepsilon > 0$ centered at $v \in \mathcal{V}(\mathcal{T})$, and let $\mathrm{Lk}(v, K(I)) \coloneqq K(I) \cap S_\varepsilon^{N-1}(v)$ denote the spherical link. Then:*

$$\Delta\chi(Q(v)) = \begin{cases} 1 - \chi(\mathrm{Lk}(v, K(I))) & \text{if } Q(v) \neq \mathbf{0}, \\ 0 & \text{if } Q(v) = \mathbf{0}. \end{cases} \tag{34}$$

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7 Specialization to 3-Dimensional Volume Processing on $\mathbb{Z}^3$

Let $\mathcal{T} = \mathbb{Z}^3$ be the standard 3D cubic lattice. Each voxel is a unit cube $\sigma_3 = [x, x+1] \times [y, y+1] \times [z, z+1]$.

7.1 Combinatorial Parameters and Denominator Clearing

- $V_0 = 1$ (1 vertex per 0-cell).
- $V_1 = 2$ (2 vertices per unit edge).
- $V_2 = 4$ (4 vertices per square face).
- $V_3 = 8$ (8 vertices per cube voxel).
- At every grid vertex $v \in \mathbb{Z}^3$, exactly $D_v = 2^3 = 8$ voxels meet.

The local configuration $Q(v) \in \{0,1\}^{2\times2\times2}$ is a $2 \times 2 \times 2$ **bit octocube** ($2^8 = 256$ local binary states).

Corollary 7.1 (Universal Hypercubic Normalization)**.** *Multiplying by $\mathrm{lcm}(V_0, V_1, V_2, V_3) = 8$ clears all fractional denominators, yielding the exact integer formula:*

$$8\chi(K(I)) = \sum_{v\in\mathbb{Z}^3} \left(8\Delta n_0(Q(v)) - 4n_1^{loc}(Q(v)) + 2n_2^{loc}(Q(v)) - n_3^{loc}(Q(v))\right). \tag{35}$$

7.2 Complete Classification of the 22 Octocube Equivalence Classes

Under the full octahedral symmetry group O_h of the cube (order 48), the 256 bit octocube configurations partition into exactly **22 equivalence classes**.

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Table 2: Classification of all 256 octocube configurations under O_h symmetry into 22 equivalence classes, with local cell allocations and integer valuations $8\Delta\chi(Q)$.

Class ID	**Voxel Count** n_3	**Multiplicity**	n_2^{loc}	n_1^{loc}	n_0^{loc} (26-conn)	$8\Delta\chi(Q)$
C_0	0	1	0	0	0	0
C_1	1	8	3	3	1	$8(1) - 4(3) + 2(3) - 1 = +1$
C_2	2 (face-adj)	12	5	5	1	$8(1) - 4(5) + 2(5) - 2 = 0$
C_3	2 (edge-adj)	12	6	5	1	$8(1) - 4(5) + 2(6) - 2 = +2$
C_4	2 (body-diag)	4	6	6	1	$8(1) - 4(6) + 2(6) - 2 = +3$
C_5	3 (corner)	24	7	6	1	$8(1) - 4(6) + 2(7) - 3 = +1$
C_6	3 (inline L)	24	8	7	1	$8(1) - 4(7) + 2(8) - 3 = +3$
C_7	3 (skew)	8	9	8	1	$8(1) - 4(8) + 2(9) - 3 = +5$
C_8	4 (square face)	6	8	6	1	$8(1) - 4(6) + 2(8) - 4 = 0$
C_9	4 (tripod)	24	9	7	1	$8(1) - 4(7) + 2(9) - 4 = +2$
C_{10}	4 (chair)	24	10	8	1	$8(1) - 4(8) + 2(10) - 4 = +4$
C_{11}	4 (skew ring)	12	10	8	1	$8(1) - 4(8) + 2(10) - 4 = +4$
C_{12}	4 (tetrahedral)	2	12	10	1	$8(1) - 4(10) + 2(12) - 4 = +8$
C_{13}	5 (compl C_5)	24	10	7	1	$8(1) - 4(7) + 2(10) - 5 = +1$
C_{14}	5 (compl C_6)	24	11	8	1	$8(1) - 4(8) + 2(11) - 5 = +3$
C_{15}	5 (compl C_7)	8	12	9	1	$8(1) - 4(9) + 2(12) - 5 = +5$
C_{16}	6 (compl C_2)	12	11	7	1	$8(1) - 4(7) + 2(11) - 6 = 0$
C_{17}	6 (compl C_3)	12	12	8	1	$8(1) - 4(8) + 2(12) - 6 = +2$
C_{18}	6 (compl C_4)	4	12	8	1	$8(1) - 4(8) + 2(12) - 6 = +2$
C_{19}	7 (compl C_1)	8	12	6	1	$8(1) - 4(6) + 2(12) - 7 = -1$
C_{20}	8 (full cube)	1	12	6	1	$8(1) - 4(6) + 2(12) - 8 = 0$
C_{21}	4 (diagonal pair)	12	10	9	1	$8(1) - 4(9) + 2(10) - 4 = +6$

8 Theory of Local Computability for the Euler Characteristic

8.1 The Local Property Counter (LPC) Formalism

To formally characterize the complexity of local counting algorithms, Gray defined the **Local Property Counter** (LPC).

Definition 8.1 (LPC Complexity 4-Tuple)**.** A local measurement algorithm Φ on an N-dimensional lattice Λ is characterized by the tuple:

$$\mathcal{L}(\Phi) \coloneqq (D, G, R, M) \in \mathbb{Z}_+^4, \tag{36}$$

where:

- **Diameter (D)**: The maximum spatial coordinate span $\max(I_1, \ldots, I_N)$ of the local neighborhood window.
- **Degree (G)**: The number of distinct Boolean logic predicates $\{F_1, \ldots, F_G\}$ feeding the property accumulator.
- **Order (R)**: The maximum number of input bit cells feeding any single Boolean predicate: $R \coloneqq \max_g |\operatorname{supp}(F_g)|$.
- **Dimension (M)**: The maximum number of orthogonal coordinate axes spanned by any single predicate F_g.

8.2 Fundamental Complexity Theorems for Local Euler Counting

Theorem 8.2 (Universal Diameter-2 Sufficiency)**.** *On any hypercubic lattice $\mathbb{Z}^N$, the topological Euler characteristic χ is computable with diameter:*

$$D = 2. \tag{37}$$

Proof. By Theorem 6.8, $\chi(K(I)) = \sum_{v \in \mathbb{Z}^N} \Delta\chi(Q(v))$. The local configuration $Q(v)$ consists of the 2^N hypercubes sharing the primal vertex v, which spans exactly a $2 \times 2 \times \cdots \times 2$ window. The diameter is $\max(2, \ldots, 2) = 2$. □

Theorem 8.3 (Order Complexity Comparison)**.** *1. For **directional sweep / critical point formulas**, the Euler characteristic has order:*

$$R = N + 1. \tag{38}$$

*2. For **isotropic dual Voronoi cell valuations**, the Euler characteristic has order:*

$$R = 2^N. \tag{39}$$

Proof. A directional Morse sweep requires sampling N independent coordinate rays relative to a base point, which requires $N + 1$ points in general position. Fully symmetric isotropic valuations require sampling all 2^N hypercube quadrants meeting at v. □

8.3 Dimensional Scaling and Master Comparison Table

Table 3: Master comparison of local Euler characteristic computation across dimensions and lattices.

Lattice / Complex	**Algorithm Type**	**Local State Space**	D	G	R	M
1D Line $\mathbb{Z}^1$	End-point sweep	$2^2 = 4$	2	1	2	1
2D Hexagonal $(6, 3)$	Critical Point	$2^3 = 8$	2	2	3	2
2D Hexagonal $(6, 3)$	Isotropic $(T_1 - T_2)$	$2^3 = 8$	2	2	3	2
2D Square $(4, 4)$	Critical Point	$2^3 = 8$	2	2	3	2
2D Square $(4, 4)$	Isotropic (W, Z, H)	$2^4 = 16$	2	3	4	2
2D Triangular $(3, 6)$	Isotropic	$2^6 = 64$	2	6	6	2
3D Cubic $\mathbb{Z}^3$	Critical Point	$2^4 = 16$	2	4	4	3
3D Cubic $\mathbb{Z}^3$	Isotropic (22 O_h classes)	$2^8 = 256$	2	22	8	3
ND Hypercubic $\mathbb{Z}^N$	Directional Morse sweep	2^{N+1}	2	$N + 1$	$N + 1$	N
ND Hypercubic $\mathbb{Z}^N$	Dual Voronoi valuation	2^{2^N}	2	$\lvert\mathcal{O}(2^N)\rvert$	2^N	N

—

9 Conclusion

In this paper, we have formalized and extended Stephen B. Gray's 1971 cell-group methodology for computing the Euler characteristic. By interpreting cell groups as dual Voronoi cells centered at primal vertices, we have proven:

1. The global Euler characteristic decomposes into a sum of local fractional valuations $\Delta\chi(Q(v)) = \sum_{k=0}^{N} (-1)^k \frac{1}{V_k} n_k^{\text{loc}}(Q(v))$, which clears denominators via the universal factor 2^N on hypercubic lattices.

2. The linear separability of cell groups explains why the hexagonal lattice is universally unambiguous (0% inseparability), while the square lattice exhibits diagonal ambiguities (12.5% inseparability).

3. The Euler differential $E^*(U) = \Delta E / \Delta U$ locally classifies simple points, enabling topology-preserving parallel skeletonization via subfield partitioning.

4. Local Euler counting is universally computable with window diameter $D = 2$ across all dimensions N, providing a sharp characterization of local computability in discrete geometry.

References

[1] S. B. Gray, "Local properties of binary images in two dimensions," *IEEE Transactions on Computers*, vol. C-20, no. 5, pp. 551–561, May 1971.

[2] M. Minsky and S. Papert, *Perceptrons: An Introduction to Computational Geometry*. Cambridge, MA: MIT Press, 1969.

[3] H. Hadwiger, *Vorlesungen über Inhalt, Oberfläche und Isoperimetrie*. Berlin: Springer-Verlag, 1957.

[4] R. Bott, *Lectures on Morse Theory*. Bonn: Bonn University Mathematical Institute, 1958.

[5] J. Milnor, *Morse Theory*. Princeton, NJ: Princeton University Press, 1963.

[6] R. Forman, "Morse theory for cell complexes," *Advances in Mathematics*, vol. 134, no. 1, pp. 90–145, 1998.

[7] A. Rosenfeld, "Connectivity in digital pictures," *Journal of the ACM*, vol. 17, no. 1, pp. 146–160, 1970.

[8] T. Y. Kong and A. Rosenfeld, "Digital topology: Introduction and survey," *Computer Vision, Graphics, and Image Processing*, vol. 48, no. 3, pp. 357–393, 1989.

[9] M. J. E. Golay, "Hexagonal parallel pattern transformations," *IEEE Transactions on Computers*, vol. C-18, no. 8, pp. 733–740, Aug. 1969.

[10] H. S. M. Coxeter, *Introduction to Geometry*. New York: Wiley, 1961.